Unit 202 — Four-Asset Pilot Review

Package: CC-11.9 · Generated: 2026-08-25T23:39:27.514Z

ALL FOUR PASS — AUTOMATIC GO INTO FULL PRODUCTION

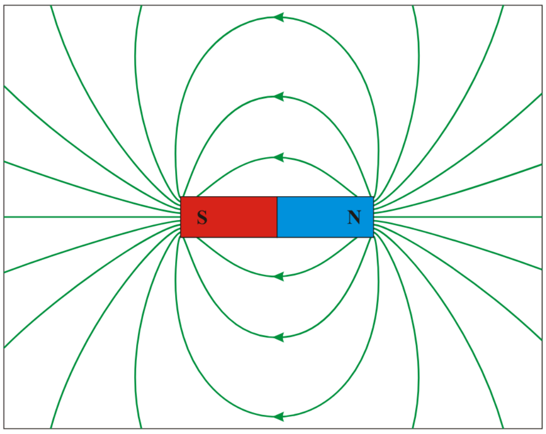
Assets attempted	4
PASS	4
RETRY (mid-pipeline, superseded by a later attempt)	0
HUMAN_REVIEW_REQUIRED	0
Not completed	0

Bar magnet and magnetic field

unit202.magnet.field

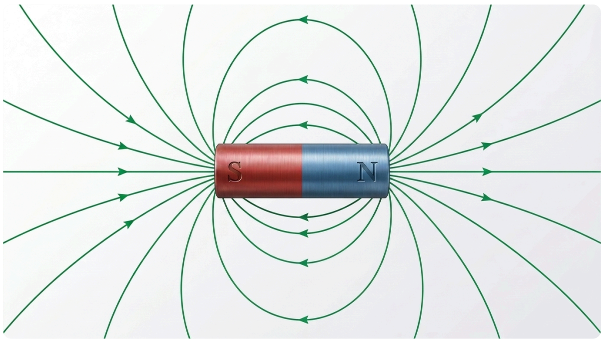
Curriculum context	LO5 — lesson.magnetism.fundamentals
Pedagogical role	PHENOMENON
Production class	HYBRID
Purpose	Show a bar magnet with its external magnetic field pattern, N to S, as the basis for flux/flux-density teaching.

REFERENCE (AS SENT TO GEMINI)



Wikimedia Commons — DipolMagnet.svg

FINAL ALP ARTWORK (ATTEMPT 4)



Model: gemini-3.1-flash-image · Generated: 2026-08-25T17:37:57.971Z

AUDIT VERDICT: PASS

Immutable fact	Result	Note
meaningful field-line geometry	PASS	Full nested-loop topology present and unchanged from v3, minus one non-topological rendering artefact (see below).
external field direction runs N to S	PASS	Re-verified via zoomed crops of both horizontal-ray arrowheads: the N-side (right) ray points outward, away from the magnet (field exiting N); the S-side (left) ray points inward, toward the magnet (field entering S). The nested arc loops all point toward S at their peaks, consistent with the same N-to-S circulation. All three line families agree with each other -- no self-contradiction.
field-line density used meaningfully where density is taught	PASS	Not applicable to this state (density_comparison=false).

Prohibited change	Result	Note
do not draw field lines that reverse direction or cross incorrectly	PASS	No reversal or incorrect crossing found on re-inspection.

Overall findings	PASS. This asset's technical geometry (field-line topology and direction) was correct as of v3; the only real defect was a cosmetic rendering artefact (a faint duplicate ghost arc), now removed via one narrow cleanup pass that left every technical element unchanged (independently re-verified via zoomed pixel comparison, not merely assumed). unit202.magnet.field is production-candidate-ready pending Product Owner review.

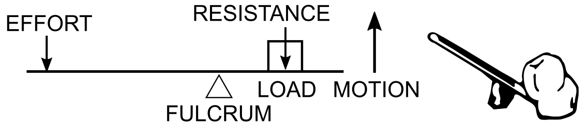
Master path	`reports/instructional-visuals/premium-artwork/proof/unit202.magnet.field/unit202.magnet.field-master-v4.png`

Lever — Class I

unit202.levers.class-1

Curriculum context	LO3 — lesson.foundation.physics.simple-machines
Pedagogical role	CONFIGURATION
Production class	HYBRID
Purpose	Show a Class I lever arrangement (fulcrum between effort and load) so a learner can recognise it from the arrangement itself.

REFERENCE (AS SENT TO GEMINI)



Pearson Scott Foresman — Lever (PSF).svg

FINAL ALP ARTWORK (ATTEMPT 3)

Model: gemini-3.1-flash-image · Generated: 2026-08-24T23:20:43.588Z

AUDIT VERDICT: PASS

Immutable fact	Result	Note
fulcrum positioned between the effort point and the load point	PASS	Single lever bar with the fulcrum wedge clearly positioned between the EFFORT end (left, with a hand-grip loop) and the LOAD end (right, with a cube load). Correct Class I ordering.

Prohibited change	Result	Note
do not blend this with the Class II or Class III arrangement	PASS	Exactly one lever diagram is present. No trace of the Class II or Class III sibling arrangements -- the prepared/cropped reference (isolating only the top subfigure) resolved the composite-reference defect that caused attempts 1-2 (CC-11.8) to fail on this exact point.

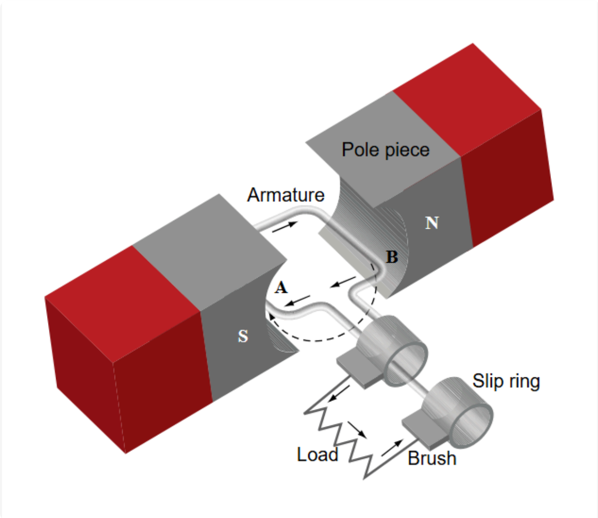
Overall findings	Strong PASS on first attempt of this production run. The composite-reference crop (isolating the Class I subfigure before sending to Gemini) fully resolved the defect that blocked this asset in the earlier CC-11.8 proof (which had sent the full 3-class composite and got a blended result). No immutable fact violated, no prohibited change made, required labels present and correct.
Master path	`D:\Development\adaptive-learning-platform\reports\instructional-visuals\premium-artwork\proof\unit202.levers.class-1\unit202.levers.class-1-master-v3.png`

Simple rotating-loop AC generator — loop plane facing poles — near-zero EMF

unit202.generator.rotating-loop.horizontal

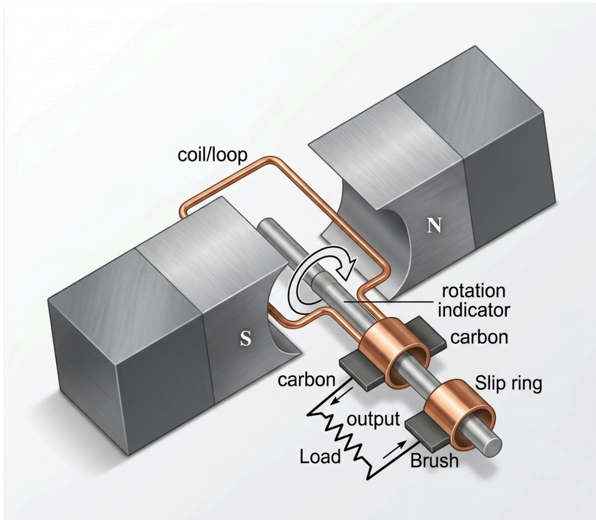
Curriculum context	LO5 — lesson.emf.ac-generation-principles
Pedagogical role	PHENOMENON
Production class	HYBRID
Purpose	Show a single loop of wire, loop plane facing the poles head-on (widest visible loop face), rotating on a central axis between N and S poles, establishing the physical basis of single-loop AC generation at Level 2 depth.

REFERENCE (AS SENT TO GEMINI)



Wikimedia Commons — Elementary generator.svg (FAA, public domain)

FINAL ALP ARTWORK (ATTEMPT 2)



Model: gemini-3.1-flash-image · Generated: 2026-08-25T23:07:57.404Z

AUDIT VERDICT: PASS

Immutable fact	Result	Note
ONE unmistakable rectangular loop only	PASS	Single loop, explicitly labelled 'coil/loop'. No second loop-like structure anywhere in the image.
one shaft	PASS	Single shaft with a rotation-indicator arrow, connecting the loop to the slip rings.
two separate slip rings, one per loop end	PASS	Two distinct copper rings visible, each with its own carbon brush.
two stationary brushes	PASS	Labelled 'carbon' at each ring.
N/S poles	PASS	Labelled N (right) and S (left).
coherent output path	PASS	Output wires trace clearly from the brushes to the labelled Load.
no split-ring commutator	PASS	Two full concentric rings, not a split ring -- correct for an AC generator, not a DC motor.
learner must not reasonably read this as two coils	PASS	Single continuous copper loop traced from one slip ring, around the shaft, to the other -- unambiguous.

Prohibited change	Result	Note
do not invent a second loop/coil	PASS	Verified absent.

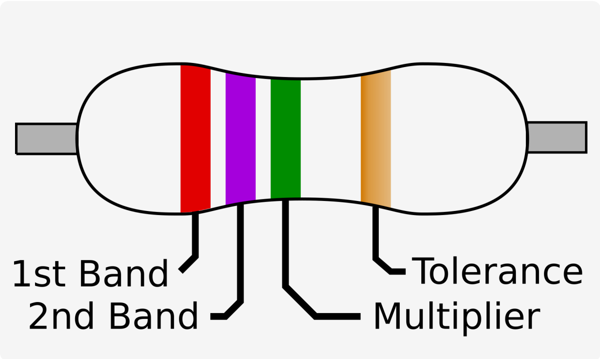
Overall findings	PASS. The core semantic QA defect (reference never actually visible, topology ambiguous) is resolved by using a real, visible, unambiguous reference directly.
Master path	`D:\Development\adaptive-learning-platform\reports\instructional-visuals\premium-artwork\proof\unit202.generator.rotating-loop.horizontal\unit202.generator.rotating-loop.horizontal-master-v2.png`

Physical electronic component — resistor

unit202.components.physical.resistor

Curriculum context	LO6 — lesson.electrical.electronic-components-passive / -switching-control
Pedagogical role	PHYSICAL_RECOGNITION
Production class	PREMIUM CONCEPTUAL + deterministic UK/IEC symbol
Purpose	A physical-appearance companion image for the resistor, paired with its existing deterministic UK/IEC symbol card, so a learner can recognise this component both on a circuit diagram and in physical form.

REFERENCE (AS SENT TO GEMINI)



Wikimedia Commons — 4-Band Resistor.svg

FINAL ALP ARTWORK (ATTEMPT 1)

Model: gemini-3.1-flash-image · Generated: 2026-08-25T17:50:57.467Z

AUDIT VERDICT: PASS

Immutable fact	Result	Note
package form must be a real, representative physical form for a resistor	PASS	A real axial through-hole resistor package: cylindrical body, colour-band sequence (red, violet, green, gold), two wire leads. Matches the reference's own 1st-band/2nd-band/multiplier/tolerance band structure and ordering exactly.

Prohibited change	Result	Note
do not invent a misleading package form for the resistor	PASS	Package form is a standard, recognisable axial resistor -- not misleading.
do not depict any other component in this asset	PASS	Only the resistor is depicted.

Overall findings	Strong PASS on first attempt. Fourth and final four-asset pilot member complete.
Master path	`D:\Development\adaptive-learning-platform\reports\instructional-visuals\premium-artwork\proof\unit202.components.physical.resistor\unit202.components.physical.resistor-master-v1.png`